

Le système international d'unités The International System of Units

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Annexe 2

Annex 2

Valeurs recommandées des fréquences étalons
destinées à la mise en pratique de la définition du
mètre et aux représentations secondaires de la
seconde

**Recommended values of standard frequencies
for applications including the practical
realization of the metre and secondary
representations of the second**

Partie 2

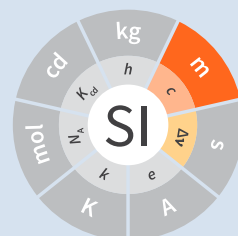
Part 2

Aluminium 27 Ion ()

Aluminium 27 Ion ()

CCL-CCTF Frequency Standards Working Group

July 2013
juillet 2013



**Recommended values of standard
frequencies for applications including
the practical realization of the metre and
secondary representations of the second
Part 2: Aluminium 27 Ion ()**

CCL-CCTF Frequency Standards Working Group

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1. Recommended value [1] of the frequency

$$f(^{27}\text{Al}^+) = 1\,121\,015\,393\,207\,857.3\text{ Hz}$$

equivalent to

$$\lambda(^{27}\text{Al}^+) = 267\,429\,385.730\,489\text{ fm},$$

with an estimated relative standard uncertainty of 1.9×10^{-15} .

This radiation was endorsed as a secondary representation of the definition of the second.

2. Source data

$$f(^{27}\text{Al}^+) = 1\,121\,015\,393\,207\,857.3\text{ Hz}$$

$$u_c/y = 1.9 \times 10^{-15}$$

calculated from

$f(^{27}\text{Al})/\text{Hz}$	u/Hz	source data
1 121 015 393 207 851	6	[2]
1 121 015 393 207 857.4	0.7	[3]

by a weighted mean.

As this value was issued from only one laboratory, the CCTF considered it prudent to estimate a relative standard uncertainty of 1.9×10^{-15} .

$^{27}\text{Al}^+$ ion, $3s^2\ ^1\text{S}_0 - 3s3p\ ^3\text{P}_0$ unperturbed optical transition

References

- [1] CIPM Recommendation 1(CI-2013): Updates to the list of standard frequencies.
- [2] T. Rosenband, P. O. Schmidt, D. B. Hume, W. M. Itano, T. M. Fortier, J. E. Stalnaker, K. Kim, S. A. Diddams, J. C. J. Koelemeij, J. C. Bergquist, and D. J. Wineland, Observation of the $^1S_0 \rightarrow ^3P_0$ Clock Transition in $^{27}\text{Al}^+$, *Phys. Rev. Lett.* **98**, 220801 (2007).
- [3] T. Rosenband, D. B. Hume, P. O. Schmidt, C. W. Chou, A. Brusch, L. Lorini, W. H. Oskay, R.E. Drullinger, T. M. Fortier, J. E. Stalnaker, S. A. Diddams, W. C. Swann, N. R. Newbury, W. M. Itano, D. J. Wineland and J. C. Bergquist: Frequency Ratio of Al^+ and Hg^+ Single-Ion Optical Clocks; Metrology at the 17th Decimal Place, *Science* **319**, 1808 (2008).